System Dynamics Society

WIP Presentation

Systems Literature Analysis Engine (SLAE)—Automating Scientific Evidence for Causal Loop Diagrams

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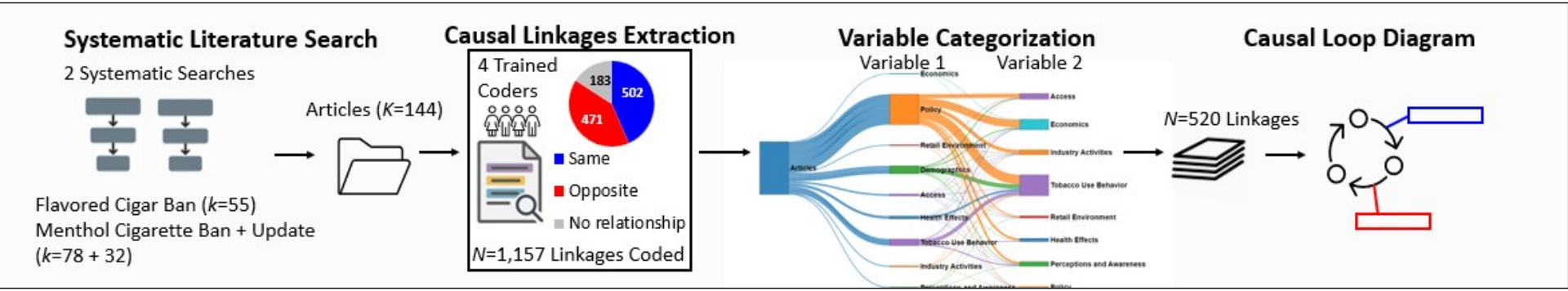
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0:00-0:30

Problem Statement

- Creating diagrams to understand the many factors in smoking control usually relies on expert knowledge or extensive research reviews.
- However, systematically reviewing literature to find these connections is a long and expensive process. Review below took 231 hours and cost $6,270.

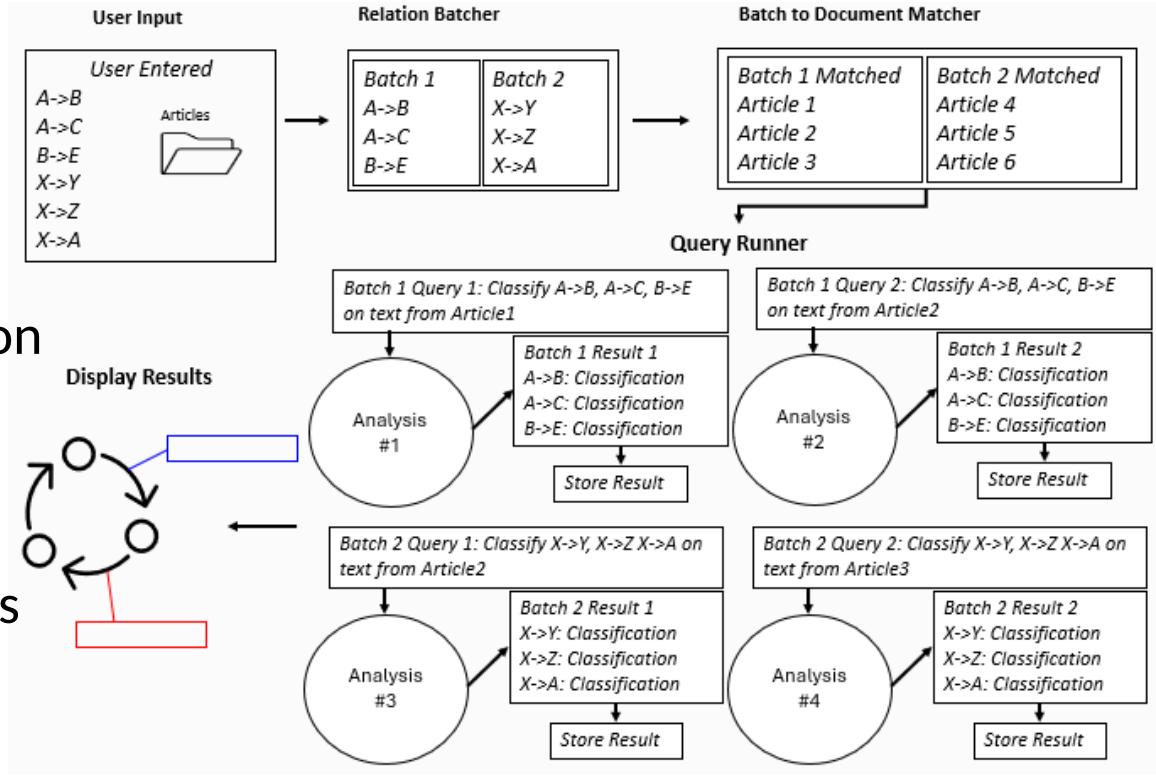


- Without a scalable, systematic method, policymakers lack confidence that causal loop diagrams reflect the full, mixed body of empirical evidence.

0:30-2:00

Approach or Dynamic Hypothesis

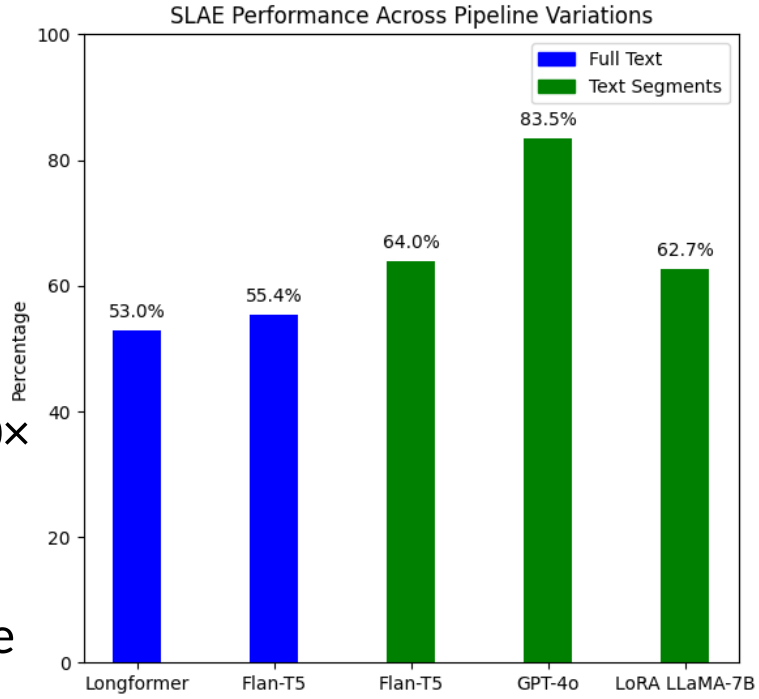
- Reviewed 144 articles on flavored tobacco bans and extracted 1,157 linkages.
- Created a coded dataset split 80/20 for training and validation of the SLAE pipeline.
- SLAE processes documents, classifies relationships, and generates causal loop diagrams with supporting evidence and polarity in Kumu.



2:00-3:30

Progress, Insights, and Questions

- **Current accuracy** 52% of polarities on average across automated approaches (2 hours).
  - Improved accuracy using a fine-tuned Longformer and confidence filter (<65%).
  - Best performance when limited to extracted text segments.
- Processing time ≈ 11 s per paper (2 h total) >100× faster than manual coding (2 h vs 231 h); cost savings ≈ $6,000.
- Next steps are to improve text extraction, reduce hallucinations, and build a web application.



3:30-5:00